

DHYEY VIPULBHAI DESAI

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PROFILE SUMMARY

AI/ML engineer building agentic AI workflows, end-to-end RAG pipelines, and LLM evaluation harnesses for enterprise clients (MS Applied Data Science, USC). Skilled in LangGraph, hybrid retrieval and reranking, multimodal ML, prompt engineering, and deployment on AWS and Spark.

EDUCATION

- University of Southern California**
Master of Science, Applied Data Science | **GPA: 3.76**
• **Coursework:** Foundations of Data Management, Machine Learning for Data Science, Foundations and Applications of Data Mining
- Manipal University Jaipur, Rajasthan**
Bachelor of Technology, Computer Science and Engineering | **GPA: 3.85**
• **Coursework:** Data Science and Machine Learning, Image Processing and Pattern Analysis, Artificial Intelligence, Regression Analysis
- Aug 2024 - May 2026**
United States of America
- Jul 2020 - May 2024**
India

PROFESSIONAL EXPERIENCE

- Associate Forward Deployed Engineer**
Momentum Blue (A Coforge Company) | Cross-Industry Enterprise AI
• **Enterprise Agentic AI Delivery Readiness:** building deployment-ready capability to design, evaluate, and deliver enterprise agentic AI.
• Built LangGraph agentic workflows and end-to-end RAG pipelines (ChromaDB, pgvector, BM25 hybrid retrieval, Cohere reranking) to Momentum Blue delivery standards, with a fixed evaluation harness for retrieval recall and answer quality.
• Ran LLM evaluations for agent and RAG quality and packaged reusable solution patterns for client deployments; a policy RAG harness scored 1.0 retrieval recall and answer accuracy on 16 questions.
- Data Science Intern**
Starcycle | LegalTech / Corporate Dissolution Platform
• Built a Slack-based RAG assistant for internal Q&A, summaries, and insights, with incremental sync, a Vue frontend, and PII redaction.
• Led CRM integration planning (API research, webhooks, field mapping); designed a staged sync that validates payloads before production.
• Built national compliance datasets with tiered state coverage, machine-readable exports, and late fee and closure reference indexes; contributed to a data reconciliation pipeline across internal and third-party systems.
- AI/ML Engineer Intern**
Onawa Pet | PetTech | [GitHub](#)
• Built a RAG system (OpenAI function calling, Firebase) over 500+ voice notes, delivering pet health analytics across 3 dashboards.
• Built a TensorFlow/Transformers pipeline for 200+ audio files (transcription, PDF analysis, summarization), cutting manual review by 60%.
- Generative AI Engineer Intern**
Genpact | IT Consulting / Digital Transformation
• Built a conversational AI assistant for purchase-order automation with multi-document querying over large purchase orders.
• Designed and optimized GPT-4 prompt sets with custom evaluation metrics for accuracy, relevance, and robustness, improving query precision.
- Aug 2026 - Present**
Princeton, New Jersey
- Feb 2026 - May 2026**
United States of America
- May 2025 - Jul 2025**
United States of America
- Apr 2024 - Jun 2024**
India

PROJECTS

- Policy RAG Assistant** | [GitHub](#)
• RAG assistant over policy documents using Gemma, Chroma, and Cohere reranking; 1.0 retrieval recall and answer accuracy on 16 questions.
- BEACON | Emergency Decision Support for First Responders** | [GitHub](#)
• QLoRA fine-tune of Gemma 4 E4B on WHO/SPHERE/IMCI protocols; offline BM25 RAG, voice/photo/text input, FastAPI on Docker.
- MultiLLM | Intelligent Multi-Model AI System**
• Privacy-first platform routing tasks across 5+ local Ollama models (Llama 3.2, DeepSeek Coder, Phi3) with streaming chat and Google OAuth.
- Sep 2025 - Dec 2025**

TECHNICAL SKILLS

- Programming Languages:** Python, SQL, Java, C, Scala | **AI Coding Tools:** Cursor, Claude, GitHub Copilot
- ML & GenAI:** LangGraph, RAG, Hybrid Retrieval (BM25), Reranking, Function Calling, QLoRA, Transformers, Recommender Systems
- LLM Evaluation:** Prompt Evaluation, Citation Checks, Fixed-Set Recall Tests | **Multimodal & NLP:** CV, NLP, NER, Text-to-SQL
- ML Engineering:** FastAPI, Pydantic, REST APIs, Docker, GitLab CI, pytest, Ruff, Mypy, Spark RDD, Hadoop, ETL, Quantization
- Tools & APIs:** Hugging Face, Ollama, TensorFlow, OpenAI/Gemini/Cohere APIs, ChromaDB, pgvector, MongoDB, Databricks, Git, Tableau
- Cloud:** AWS Certified AI Practitioner (2026), AWS Certified Cloud Practitioner (2025)

PUBLICATIONS

- Amplified Silence: Uncovering and Mitigating Gender Bias in Music Recommendation Algorithms** – accepted and presented at the 4th Music Recommender Systems Workshop (MuRS 2026), held with RecSys, Sept 2026. Stacking a post-processing boost with Fair-ALS cut male-artist exposure from 69.8% to ~50% at 30% lower NDCG cost than post-processing alone.
- Pseudo-RGB Slice Stacking in a 2D ResUNet for High-Sensitivity Multiple Sclerosis Lesion Segmentation, Diagnostics (MDPI)** ([code](#)).
- Supervised Machine Learning Approaches for Brain Stroke Detection**, IEEE IEMECON 2023: logistic regression reached 97% accuracy.

ACHIEVEMENTS

- Origin Weekend: IMPACT S26 – First Place:** won a startup launch sprint organised by USC, Google, and TIE Hub (USC Viterbi).
- Graduate Teaching Assistant, DSCI 551 & DSCI 351:** supported 100+ students via weekly sections; designed assignments with faculty.
- AWS re:Invent:** All Builders Welcome Grant recipient (2025); Alumni Advisor (2026) | **Dean's List** (2023)
- Advanced Deep Learning Research, NUS:** privacy-preserving facial analysis; +10% accuracy, 40% lower latency on 3× larger batches.